

LAMPIRAN-1: PENYELESAIAN DENGAN PROGRAM PYTHON

Bagian ini menyediakan skrip Python untuk memvalidasi hasil dan melengkapi cara mengerjakan selain dengan menggunakan Spreadsheet atau PSPP pada bab-bab sebelumnya.

Persiapan Library

Sebelum memulai, pastikan sudah terinstall python dan pustaka berikut:

1. Install software Python, Thonny, pelajari menu, menulis dan menjalankan coding python, bagaimana install library.
2. Install library dengan perintah: `pip install pandas numpy matplotlib seaborn statsmodels scipy`,
3. Dalam koding terlebih dulu import library tersebut no 2 yang sesuai dengan persoalan, dengan menulis di awal program:
 - a. `import pandas as pd`,
 - b. `import numpy as np`,
 - c. `import matplotlib.pyplot as plt`,
 - d. `import seaborn as sns`,
 - e. `import statsmodels.api as sm`,
 - f. `from scipy import stats`.

1. Contoh Soal-1: Statistik Deskripsi & Visualisasi

Kasus: Pengeluaran operasional 12 gerai ritel.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# --- Contoh 1: Statistik Deskripsi ---
data_operasional = [12, 15, 14, 13, 16, 45, 14, 15, 12, 14, 13, 15]
df1 = pd.DataFrame(data_operasional, columns=["Operasional"])
print(df1.describe()) # Menampilkan mean, std dev, dll
print("median : ", np.median(data_operasional))
plt.boxplot(data_operasional)
plt.show()
```

Contoh-1b Visualisasi dengan python:

```
import matplotlib.pyplot as plt
import seaborn as sns
# --- Contoh 2: Deteksi Outlier ---
pengeluaran = [12, 15, 14, 13, 16, 45, 14, 15, 12, 14, 13, 15]
sns.boxplot(x=pengeluaran)
plt.title('Deteksi Outlier Pengeluaran Ritel')
plt.show()
```

2. Contoh Soal-2: Korelasi

Kasus: Hubungan jam pelatihan (X) dan produktivitas (Y).

```

import pandas as pd
import seaborn as sns
# --- Contoh 3: Korelasi (Pearson) ---
data_kor = {'Jam': [10, 15, 20, 25, 30], 'Produktivitas': [60, 70, 80, 85, 95]}
df_kor = pd.DataFrame(data_kor)
print(df_kor.corr())

```

3. Contoh Soal-3: ANOVA

Kasus: Perbandingan penjualan di 3 cabang (Sukun, Lowokwaru, Klojen).

```

import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols
# --- Contoh 4: One-Way ANOVA ---
data_anova = pd.DataFrame({
    'Penjualan': [20, 22, 21, 19, 20, 25, 28, 26, 27, 29, 21, 23, 20, 22, 21],
    'Cabang': ['Sukun']*5 + ['Lowokwaru']*5 + ['Klojen']*5
})
model_anova = ols('Penjualan ~ C(Cabang)', data=data_anova).fit()
print(data_anova)
print(sm.stats.anova_lm(model_anova, typ=2))

```

twoway anova:

```

import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols
# 4. Twoway Anova
data_anova2 = {
    'Penjualan': [20, 22, 21, 15, 16, 14, 28, 30, 29, 20, 19, 21],
    'Lokasi': ['Sukun', 'Sukun', 'Sukun', 'Sukun', 'Sukun', 'Sukun',
               'Lowokwaru', 'Lowokwaru', 'Lowokwaru', 'Lowokwaru', 'Lowokwaru',
               'Lowokwaru'],
    'Media': ['Medsos', 'Medsos', 'Medsos', 'Brosur', 'Brosur', 'Brosur',
              'Medsos', 'Medsos', 'Medsos', 'Brosur', 'Brosur', 'Brosur']}
df2 = pd.DataFrame(data_anova2)
# 2. Membuat Model ANOVA (C berarti data Kategorik)
# Rumus: Penjualan dipengaruhi oleh Lokasi, Media, dan Interaksi Lokasi:Media
model = ols('Penjualan ~ C(Lokasi) + C(Media) + C(Lokasi):C(Media)', data=df2).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
# 3. Menampilkan Hasil
print("Hasil Two-Way ANOVA:")
print(df2)
print(anova_table)
# 4. Visualisasi Interaksi (Opsional namun sangat disarankan untuk Skripsi)
import seaborn as sns
import matplotlib.pyplot as plt

```

```
plt.figure(figsize=(8,5))
sns.pointplot(data=df2, x='Lokasi', y='Penjualan', hue='Media', markers=["o", "s"],
linestyles=["-", "--"])
plt.title('Grafik Interaksi Lokasi dan Media Promosi')
plt.show()
```

4. Contoh Soal-4: Regresi Linear Sederhana

```
import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols
x = sm.add_constant([2, 4, 6, 8, 10]) # Menambah konstanta (intercept)
y = [10, 15, 25, 30, 40]
model = sm.OLS(y, x).fit()
print(model.summary())
```

5. Contoh Soal-5 & 6: Regresi Multilinear & Uji Parsial/Serentak

```
import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols
data = {'Penjualan': [50, 60, 70, 80, 90],
        'Iklan': [5, 7, 8, 10, 12],
        'Sales': [2, 3, 3, 4, 5]}
df_multi = pd.DataFrame(data)
X = df_multi[['Iklan', 'Sales']]
X = sm.add_constant(X)
Y = df_multi['Penjualan']
model_multi = sm.OLS(Y, X).fit()
print(model_multi.summary()) # Menampilkan Uji F (serentak) dan Uji t (parsial)
```

6. Contoh Soal-7 & 8: Prediksi (Forecasting)

```
import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols
data = {'Penjualan': [50, 60, 70, 80, 90],
        'Iklan': [5, 7, 8, 10, 12],
        'Sales': [2, 3, 3, 4, 5]}
df_multi = pd.DataFrame(data)
X = df_multi[['Iklan', 'Sales']]
X = sm.add_constant(X)
Y = df_multi['Penjualan']
model_multi = sm.OLS(Y, X).fit()
print(model_multi.summary())
# Menggunakan model yang dihasilkan untuk prediksi baru
# Iklan = 10, Sales = 4
```

```
X_baru = [1, 10, 4] # 1 adalah konstanta
prediksi = model_multi.predict(X_baru)
print(f"Hasil Prediksi Penjualan: {prediksi[0]:.2f}")
```

7. Contoh Soal-9: Uji Asumsi (Multikolinearitas)

```
import pandas as pd
import statsmodels.api as sm
from statsmodels.formula.api import ols
from statsmodels.stats.outliers_influence import variance_inflation_factor
# --- Contoh 8 & 9: Uji Multikolinearitas (VIF) ---
data_multi = {'Y': [50, 60, 70, 80, 90], 'X1': [5, 7, 8, 10, 12], 'X2': [2, 3, 3, 4, 5]}
df_m = pd.DataFrame(data_multi)
X_m = sm.add_constant(df_m[['X1', 'X2']])
vif_data = pd.DataFrame()
vif_data["Variabel"] = X_m.columns
vif_data["VIF"] = [variance_inflation_factor(X_m.values, i) for i in
range(len(X_m.columns))]
print(vif_data)
```

8. Contoh Soal-10: Hipotesis (T-Test)

```
import statsmodels.api as sm
from statsmodels.formula.api import ols
from scipy import stats
# --- Contoh 10: One Sample T-Test ---
pengunjung = [45, 48, 44, 46, 45, 47, 44, 45, 46, 45]
t_stat, p_val = stats.ttest_1samp(pengunjung, 50)
print(f"P-Value: {p_val}")
```